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Reproduction, recruitment and fragmentation in nine sympatric species of the coral genus *Acropora*

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Abstract

Multispecies assemblages of the coral genus *Acropora* occur commonly throughout the Indo-Pacific Ocean. Nine species from such an assemblage comprising 41 species of *Acropora*, at Big Broadhurst Reef on the Great Barrier Reef, were studied during 1981–1983. Similarities and differences in reproductive modes and timing, oocyte dimensions and fecundity, recruitment by larvae and by fragments, and mortality were recorded. All species had an annual gametogenic cycle, were simultaneous hermaphrodites, and had the same arrangement of gonads in polyps. In six species, most colonies released gametes on the same night of the year, in early summer, during a mass spawning event involving many coral genera. A seventh species had colonies spawning at this as well as other times of the year. Another species spawned in late summer, and gametes were not observed to mature in the last species. Eggs were very large (601 to 728 μm geometric mean diameter) and fecundity of polyps low, compared with other corals; no reduction in oocyte numbers occurred during oogenesis. Reef-flat species had slightly bigger and fewer eggs than reef-slope species. All species recruited by larvae, but four also multiplied by fragmentation, either year-round or during occasional rough weather. Year-round fragmenters had few larval recruits; non-fragmenters had many, and a rough-weather fragmenter had an intermediate number of larval recruits. It was concluded that larval recruitment largely determined species composition, and that reduced larval recruitment was responsible for sparse distribution of fragmenting species. Subsequent mortality in some species and increase by fragmentation in others probably determined relative abundances.

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Introduction

Loya (1983) suggested that reproductive isolation may be a factor contributing to species diversity in the Red Sea, since species there partition their times of spawning. Many other explanations for maintenance of diversity in coral assemblages have been explored (e.g. see Connell, 1978; Sheppard, 1982), but these can not be fully elucidated, nor can the hypothesis of reproductive isolation be tested elsewhere, while life-history characteristics of the participating species are unknown. The present study examines life-history parameters, particularly those relating to reproduction and recruitment, for nine sympatric species of the coral genus *Acropora* occurring within a high-diversity association on a reef front.

In the Great Barrier Reef Province, the coral genus *Acropora* has some 73 species (Veron and Wallace, 1984), of which 41 coexist within a section of reef front 10 m wide by 30 m on the reef studied (Wallace, 1979; and additional records in Veron and Wallace, 1984). Assemblages of *Acropora* are characteristic of mid- to outer-continental shelf reefs in eastern Australia (Done, 1982) and are the main feature of the coral fauna distinguishing these offshore reefs from inshore reefs, which lack a significant *Acropora* assemblage. From a recruitment and transplantation study, Sammarco (1983) concluded that greater recruitment of *Acropora* species in the offshore reefs and greater mortality of juvenile *Acropora* spp. in inshore reefs both contributed to this distribution.

On a broader scale, an *Acropora* spp. assemblage is a widespread and long-recognised feature of Indo-Pacific reefs (e.g. Darwin, 1842; Mayor, 1924; Wells, 1954; Rosen, 1971; Randall, 1973; Mergner and Scheer, 1974; Morton, 1974), and its location relative to other coral assemblages on these reefs is apparently correlated with strong water movement and good light penetration (Randall, 1973; Rosen, 1975; Pichon, 1978; Done, 1982).

Several recorded characteristics of *Acropora* suggest an early successional role for this genus. It has been seen to

Table 1. *Acropora* spp. Features of species studied and their distribution at the sites on Big Broadhurst Reef in October 1980, from three quadrats, each 25 m², at each site

Species	Colony shape[a]	Geographic range[a]	Total no. of colonies (no. of quadrats)			
			Inner flat	Outer flat	Buttresses	Channels
A. loripes	Caespitose, corymbose	Central Indo-Pacific (Indonesia to Great Barrier Reef)	0	0	31 (3)	5 (2)
A. granulosa	Plate		0	0	17 (3)	2 (1)
A. sarmentosa	Bottlebrush, plate		4 (2)	1 (1)	10 (3)	7 (3)
A. longicyathus	Bottlebrush		0	0	19 (3)	17 (2)
A. florida	Bottlebrush, arborescent	Broad Indo-Pacific (Andaman Sea to Marshall Islands)	0	0	6 (2)	18 (2)
A. horrida	Arborescent, bottlebrush, caespitose		0	0	5 (2)	3 (3)
A. nobilis	Arborescent		5 (2)	127 (3)	0	24 (2)
A. hyacinthus	Plates, tables	Cosmopolitan Indo-Pacific, except Hawaii and W. America	24 (3)	240 (3)	1 (1)	2 (1)
A. valida	Corymbose, caespitose	Cosmopolitan Indo-Pacific, including Hawaii and Colombia[b]	7 (2)	143 (3)	0	0

[a] Data from Veron and Wallace (1984)
[b] Data from J. W. Wells (personal communication)

dominate coral assemblages recovering from damage, particularly after predation by *Acanthaster planci* (Pearson, 1981) and to form the major taxonomic component of recruits to settlement plates in many localities (Bothwell, 1982; Sammarco and Carleton, 1982; Wallace and Bull, 1982; Wallace, 1983 a). It also has a rapid growth rate relative to that of other genera (Mayor, 1924; Randall, 1973; Buddemeier and Kinsey, 1976), and many of its species regenerate from fragments (Bak and Criens, 1982; Bothwell, 1982; Highsmith, 1982). These assessments, however, do not apply to all species, and there may well be, within *Acropora* assemblages, species which recruit poorly from larvae, grow slowly, or fail to reproduce asexually by fragmentation.

The nine species studied here cover a range of colony forms from thin plates and low clumps to bottlebrush- and open branching-forms (Table 1), suggesting a range of susceptibilities to asexual reproduction by fragmentation (Randall, 1973; Collins, 1978; Heyward, 1981; Tunnicliffe, 1981; Bothwell, 1982; Highsmith, 1982). At the outset of the study, nothing was known of their reproductive phenologies, spawned products, sexuality or fecundity. All species belong to subgenus *Acropora* (*Acropora*) Veron and Wallace, 1984, in which both gametic spawning (Bothwell, 1982) and viviparity (Stimson, 1978) have been reported. In the three species of another subgenus, *Acropora* (*Isopora*) Studer, 1878, only viviparity has been reported (Atoda, 1951; Bothwell, 1982; Veron and Wallace, 1984; Kojis, personal communication).

The assemblage in which these species occur has been described elsewhere from line transects (Wallace, 1979; Wallace and Bull, 1982). There are 40 genera of scleractinian corals on the reef front and 41 species of *Acropora*. Five of the species occur only below low water (*Acropora loripes*, *A. granulosa*, *A. longicyathus*, *A. florida* and *A. horrida*). One (*A. sarmentosa*) occurs above and below low water, but is never common. Two (*A. nobilis* and *A. hyacinthus*) are very common species, occurring mainly around and just below low water, and *A. valida* is moderately common around low water.

Materials and methods

Sites

Study sites were on Big Broadhurst Reef, a mid-shelf patch-reef north-east of Townsville, East Australia

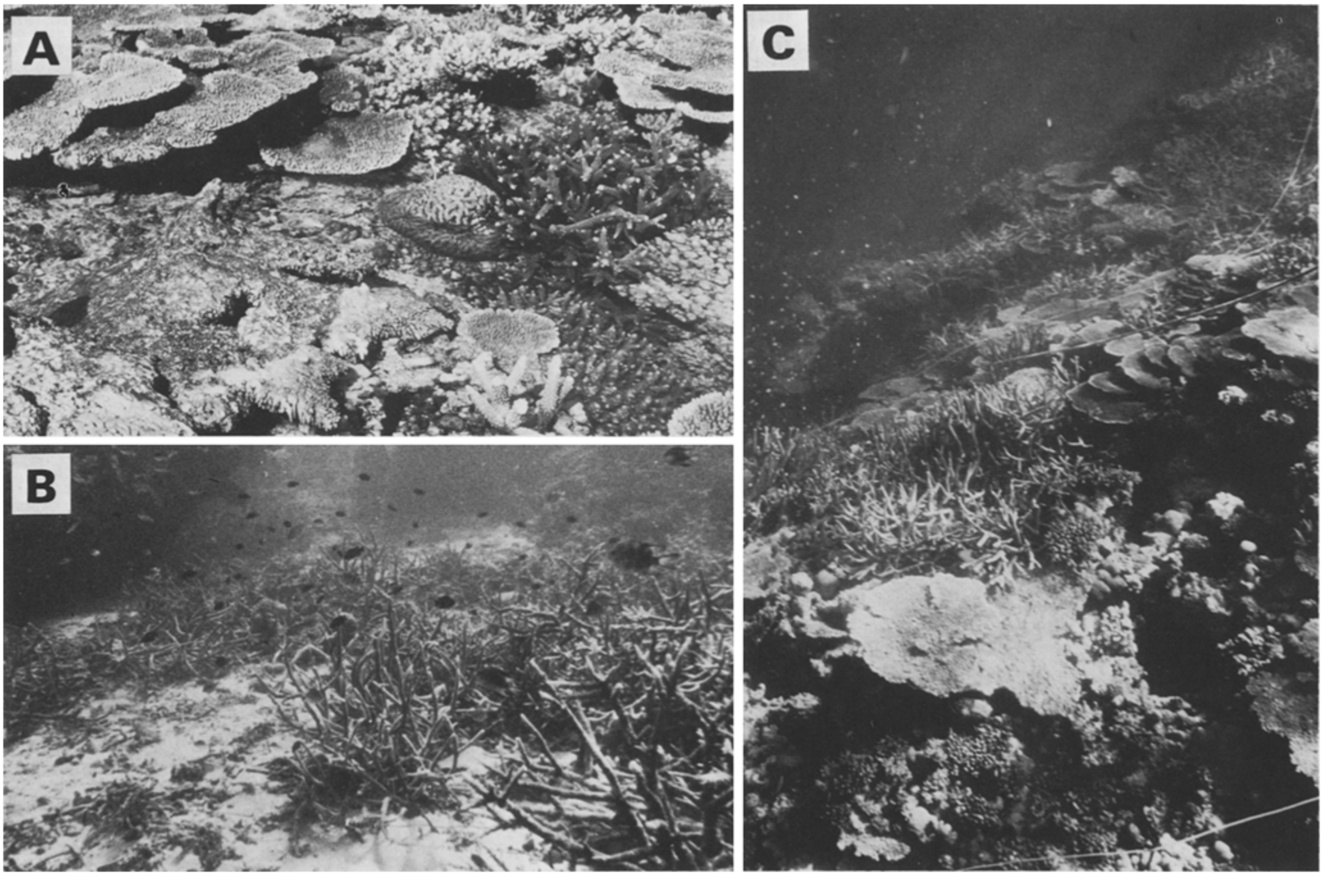

Fig. 1. The three types of adjacent *Acropora* spp.-dominated reef-front habitats sampled. (A) Outer reef flat, at tidal datum, only habitat (with inner flat) of *A. valida* (caespitose), main habitat of *A. nobilis* (arborescent) and *A. hyacinthus* (plates); (B) surge channels with gently sloping sandy floors, at approx 15 m, main habitat for *A. horrida* (arborescent), with *A. longicyathus* (bottlebrush) and *A. nobilis* (arborescent) also common; (C) reef-front buttresses, flanking surge channels, sites centered at 10 m, main habitat of all other species

(18°55′S; 147°44′E: see map in Wallace and Dale, 1977). Along the south-west side of this reef a condition known as "false front" occurs, where a channel approximately 25 m wide and 20 m deep separates two parallel strips of reef flat. The outer reef front in this area is formed into buttresses 15 to 20 m wide, alternating with surge channels opening to similar widths. Substratum on the reef flats and buttresses is solid, that in the surge channels is fine sediment and rubble. Species of *Acropora* dominate the outer reef flat, the slopes of buttresses to about 15 m, and the surge channel floors (Wallace and Bull, 1982). Studies of reproduction, recruitment and mortality were made on the buttresses and in the surge channels (Fig. 1 B, C). Studies of reproduction were also made on the outer reef flat (Fig. 1 A).

In each of four habitats (inner and outer reef flats, buttresses and channels), three quadrats, 5 m × 5 m, were marked. Those on the reef flats were randomly situated approximately 15 m apart. Those on the buttresses were centered at 10 m depth on three buttresses of similar physical structure (aspect and slope), while surge channel quadrats were at 15 m depth on the floor of three surge channels. All quadrats were within a length of reef front of 300 m. Initial population size within quadrats was assessed for each species, and a list was made of all other species of *Acropora* present.

Studies of recruitment and mortality

Recruitment and mortality were studied in the three buttress and three channel quadrats from February 1981 to February 1983. (A similar study was also commenced on the outer flat, but this was discontinued after most labels were removed by physical causes or grazed by fish, or else obliterated by the rapid growth of undifferentiated coenosteum, which only occurred in this habitat.)

All colonies of the study species in the six quadrats were tagged with impressed plastic labels. An estimate of size of each colony was made by measuring its length and breadth in a horizontal plane, and its position in the quadrat was mapped by compass-and-line bearings. Tagged colonies were 4 cm or more in diameter, this being the smallest size at which colonies could be easily recognised. In subsequent visits four months apart (with occasional additional visits), divers searched for all tags and noted whether each colony was alive or dead. Where possible, cause of death was recorded. Any newly appearing colonies were tagged, measured and mapped, and recorded as either a newly visible recruit or a fragment. The two were easily distinguished, as fragments were asymmetrical and either unattached or asymmetrically attached to the substratum, while larval recruits had a symmetrical basal area from which rose incipient branches.

In addition to the tagging program, a study of larval recruitment was made in October 1982. From other studies (Wallace, 1983 b), it was known that juveniles recruited from larvae since October 1980 would be 5 cm or less in diameter by this time, and could be located by close scrutiny of the substratum. Since this excercise required specialist knowledge of the identification of extremely small colonies of *Acropora* spp., and took 12 h diving time, it was done only once during the study. Quadrats were divided corner-wise into four segments and recruits 5 cm or less in diameter were sought. For those identified as belonging to the study species, identity and maximal and median diameter were recorded.

Studies of reproduction

During the initial mapping and tagging, samples for reproductive studies were taken from at least ten colonies of each species, and on all subsequent visits these colonies were re-sampled. In this way a record was made of the gametogenic cycle for individual colonies and for populations. At some sampling times other tagged colonies were added to those sampled, to compensate for heavy mortality from *Acanthaster planci* predation.

Samples were fixed in 10% seawater Formalin for a minimum of one week, then decalcified in 4% formic acid solution and transferred to 10% Formalin. They were dissected under a binocular dissecting microscope, lit by a fibre optic light source, as follows. The polyp was removed from the branch with forceps, and placed with its inner (branch) side uppermost on a wax dissecting dish. The stomodeum was severed between the two ventral mesenteries. The opened stomodeum was then pinned to the wax with fine steel pins, and the mesenteries were gently teased out so that they formed a sequence, with ventral directive mesenteries outermost and dorsal directive mesenteries innermost (as in the dissections in Fig. 4). In this way, both the position and number of mesenteries containing developing gonads could be recorded. Maximal and median diameter of each oocyte were measured, and number of oocytes in the polyp was counted. Using this method, rather than histological sections, all oocytes in a polyp, rather than just a sample of these, could be accurately counted and measured. A sample of 5 polyps per colony, or per location in the colony, was found to give statistically valid comparisons. Gonad structure was examined in histological sections stained with haemotoxilin and eosin or Gommori's trichrome.

To determine distribution of fertile polyps in colonies, large samples were taken from random colonies outside the quadrats. All branches in these specimens were slit open and examined under the binocular microscope for the presence of polyps bearing gonads.

Variation in size of mature oocytes, and in fecundity of polyps within colonies, was determined for six species (*Acropora loripes, A. sarmentosa, A. longicyathus, A. florida, A. hyacinthus* and *A. nobilis*). Specimens of these species were collected in November 1983 (four specimens per species, except three for *A. nobilis* and one for *A. sarmentosa*). Samples of five polyps were taken from branchlets, tips and bases of branches, and at intervals of 1 or 2 cm along branches.

For all species mentioned above except *Acropora sarmentosa,* polyp fecundity was compared for the years 1981, 1982 and 1983 from ten specimens randomly collected in the month before spawning each year.

Observations on spawning

A technique for predicting spawning date described by Harrison *et al.* (1984) was used. Freshly broken pieces of coral were examined microscopically. It was predicted that spawning would occur during the week following the first full moon after coloured eggs and motile sperm were present. During this week, in 1982 and 1983, dives were made at night to watch for spawning in the field. In 1982, a single night dive was made five nights after the full moon. In 1983, colonies of the study species and other species of *Acropora* were tagged and sampled to assess their reproductive status, and dives were made each night for five nights to observe spawning *in situ.* Tagged colonies were re-sampled each day to determine whether reproductive products had been expelled. Colonies were also kept in aquaria on board ship for spawning observations.

Results

Recruitment

Recruits of *Acropora* spp. into the buttress and channel quadrats were from three categories: (1) larval recruits which were detected by close scrutiny of the substratum; (2) fragments, which were either unattached or loosely attached to the substratum; (3) "apparent" recruits, i.e., juveniles with branching from a distinct basal area, which were less than 4 cm in diameter at the beginning of the study, but were recorded and tagged during the first year. The first two categories are discussed here, and the third category is included in the estimate of initial population size (Table 2).

All species showed larval recruitment (Table 2); however recruitment was predominantly or solely larval in *Acropora hyacinthus, A. granulosa, A. sarmentosa* and *A. loripes,* and predominantly by fragments in *A. longicyathus, A. horrida* and *A. nobilis. A. florida* employed both modes (Table 2). Recruitment by fragments only occurred

Table 2. *Acropora* spp. Comparison of recruitment by larvae and by fragments for each species in buttress and channel sites over 2 yr. *N*: initial population size, including juveniles > 5 cm diam tagged during first year

Species	*N*	Larval recruits				Fragments			
		Mean diam, cm (± SE)	On buttresses	In channels	Total	Mean diam, cm (± SE)	On buttresses	In channels	Total
A. loripes	36	2.8 (0.4)	10	17	27	8 (3)	2	0	2
A. granulosa	19	3.9 (0.5)	7	8	15	0	0	0	0
A. sarmentosa	17	5.4 (2.6)	4	3	7	0	0	0	0
A. longicyathus	36	2.5	0	1	1	22 (3)	34	15	49
A. florida	29	3.5 (0.9)	5	4	9	31 (6)	15	2	17
A. horrida	40	2.0 (0.2)	0	4	4	11 (2)	0	30	30
A. nobilis	24	6.0 (2.0)	0	2	2	24 (6)	0	17	17
A. hyacinthus	2	2.3 (0.3)	15	13	28	~500	1	0	1
A. valida	0	0	0	0	0	0	0	0	0

Table 3. *Acropora* spp. Mortality suffered by different categories and species in buttress and channel sites over 2 yr. (For juveniles and recruited fragments, only mortality of those tagged during first 9 mo is considered)

Species	Mature colonies (> 3 yr old)		Juveniles (2–3 yr old)		Recruited fragments	
	Mean diam, cm (± SE)	Mortality (%)	Mean diam, cm (± SE)	Mortality (%)	Mean diam, cm (± SE)	Mortality (%)
A. loripes	13 (2)	57	10 (1)	66	0	0
A. granulosa	16 (2)	33	6 (1)	33	0	0
A. sarmentosa	29 (2)	50	13 (3)	57	0	0
A. longicyathus	32 (9)	66	0	0	22 (3)	52
A. florida	36 (6)	17	18 (3)	66	31 (6)	23
A. horrida	16 (2)	74	0	0	11 (2)	53
A. nobilis	25 (4)	25	0	0	24 (6)	53

in quadrats where the species was already present, except for one *A. hyacinthus* plate which fell onto a channel site from a buttress above. *A. valida,* which was absent from these quadrats, had no recruitment (Table 2), although this species shows great larval recruitment on the reef flat (Wallace, unpublished data).

Most fragmentation recruitment by *Acropora florida* was from breakage of two large colonies on one buttress during a cyclonic disturbance. *A. longicyathus* recruited onto buttresses as fragments regularly throughout the year. Fragments of this species did not cement themselves to the substratum, and were thus very mobile. In channels, some recruitment of *A. longicyathus* was by survival of small portions of large colonies preyed upon by *Acanthaster planci. Acropora horrida,* which has easily broken slender branches, recruited regularly in the channels from small fragments.

Larval recruits appeared independently of adult colony presence (Table 2). The greatest larval recruitment was from *Acropora hyacinthus,* which was rare as mature colonies in these habitats, although dominant on the outer flat.

Mortality

The major cause of death during the study was from predation by the seastar *Acanthaster planci.* A single individual of this seastar was present in each of two buttress quadrats for 18 mo, and aggregations were present in the vicinity of the channel quadrats in the last 12 mo of the study. Both attached and unattached colonies of all species were preyed upon. Heaviest mortality occurred within the juvenile corals (Table 3), since predation on these colonies resulted in death to the whole colony. Some large colonies were preyed upon but not killed: in 16 cases, colonies bearing feeding scars survived.

Two species with poor cementing properties and therefore highly mobile colonies and fragments (*Acropora horrida* and *A. longicyathus*) suffered heavy mortality of both adults and recruited fragments (Table 3), probably due to the combined effects of *Acanthaster planci* predation and chronic movement and burial. Two colonies of *Acropora florida* and one of *A. loripes* rolled out of the quadrats.

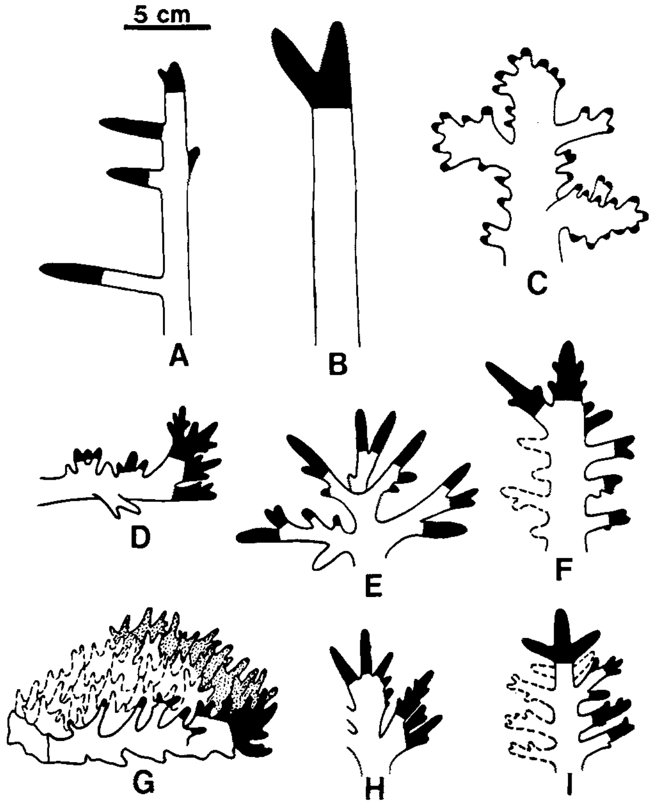

Fig. 2. *Acropora* spp. Location of "sterile zones" (regions in which polyps bear no gonads), shown in black, on representative specimens of each species, drawn to scale: (A) *A. horrida* (from upper buttress); (B) *A. nobilis* (from outer flat); (C) *A. sarmentosa*, (D) *A. granulosa*, (E) *A. loripes*, (F) *A. florida*, (I) *A. longicyathus* (all from reef buttresses); (G) *A. hyacinthus*, (H) *A. valida* (both from outer flat)

Sexual reproduction

Distribution of fertile polyps

Acropora differs from other coral genera in having axial and radial polyps, located in the centre and along the sides of branches, respectively, extension occurring at the tip of the axial corallite and new radial corallites being budded off from the extending tip. While both types of polyp were capable of developing gametes, gravid axial polyps were relatively rare (seen only in some colonies of *A. loripes*, *A. granulosa*, *A. sarmentosa*, *A. hyacinthus* and *A. valida*). Axial polyps apparently were only gravid in branches that were not actively extending. For example, samples taken at intervals inwards from the edge zone of *A. hyacinthus* plates would eventually encounter branchlets with gravid axials, at 11 to 15 cm in from the edge zone; small branchlets back from the edge zone in *A. loripes* and *A. granulosa* were sometimes fully gravid, including axial polyps (Fig. 2 D, E).

Most branch tips bore a zone of sterile radial corallites. The length of this sterile zone was extremely variable (Fig. 2) and it apparently consisted of radial corallites which had been budded after the time of onset of gametogenesis. It was entirely absent from some branchlets, whose position on the inside of the colony indicated that they were not extending (Fig. 2 D, E).

Gonad structure and location

All nine species were simultaneous hermaphrodites and had the same arrangement of gonads in the polyps.

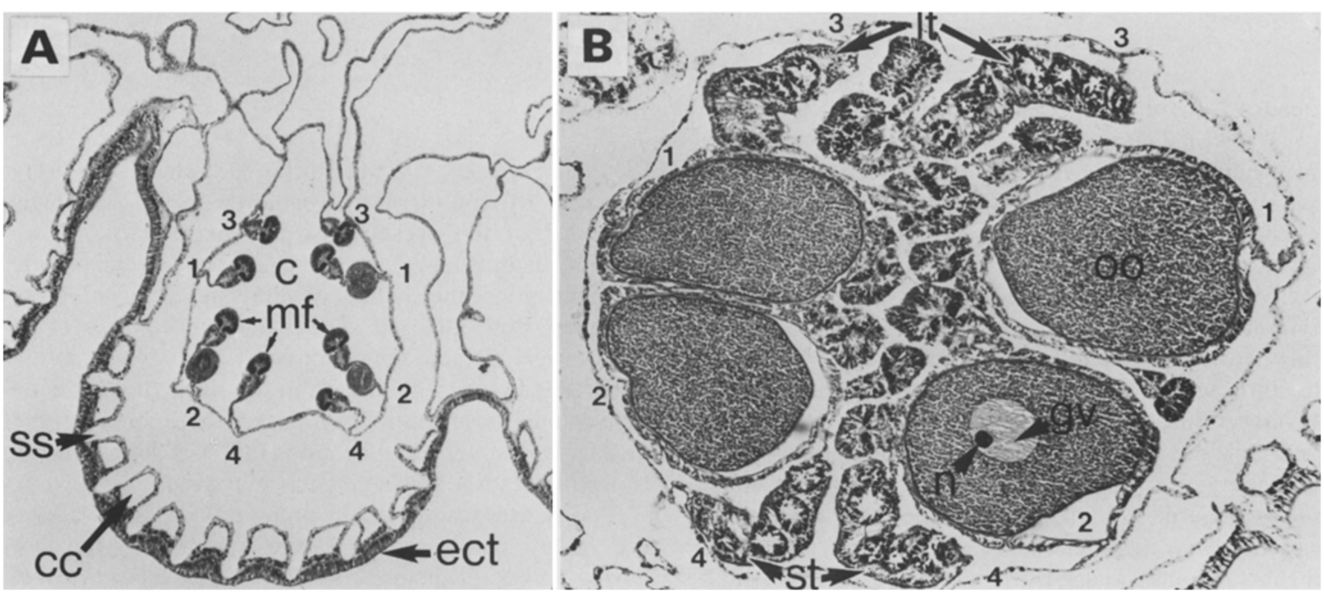

Fig. 3. *Acropora granulosa.* Histology of polyps undergoing early- and late-stage gametogenesis. (A) Transverse section of radial polyp showing position of sequentially formed pairs of mesenteries (1–4), with developing oocytes in mesoglea of Pairs 1 and 2; (B) transverse section of sexually mature radial polyp, with oocytes (oo) fully developed in Mesenterial Pairs 1 and 2, and loculated testes developed in Pairs 3 (large testes, lt) and 4 (small testes, st). cc: canaliculi: of coelenteron (c); ect: ectoderm; gv: germinal vesicle; mf: mesenterial filaments; n: nucleolus; ss: position of skeletal spinules. (A) ×25, (B) ×75

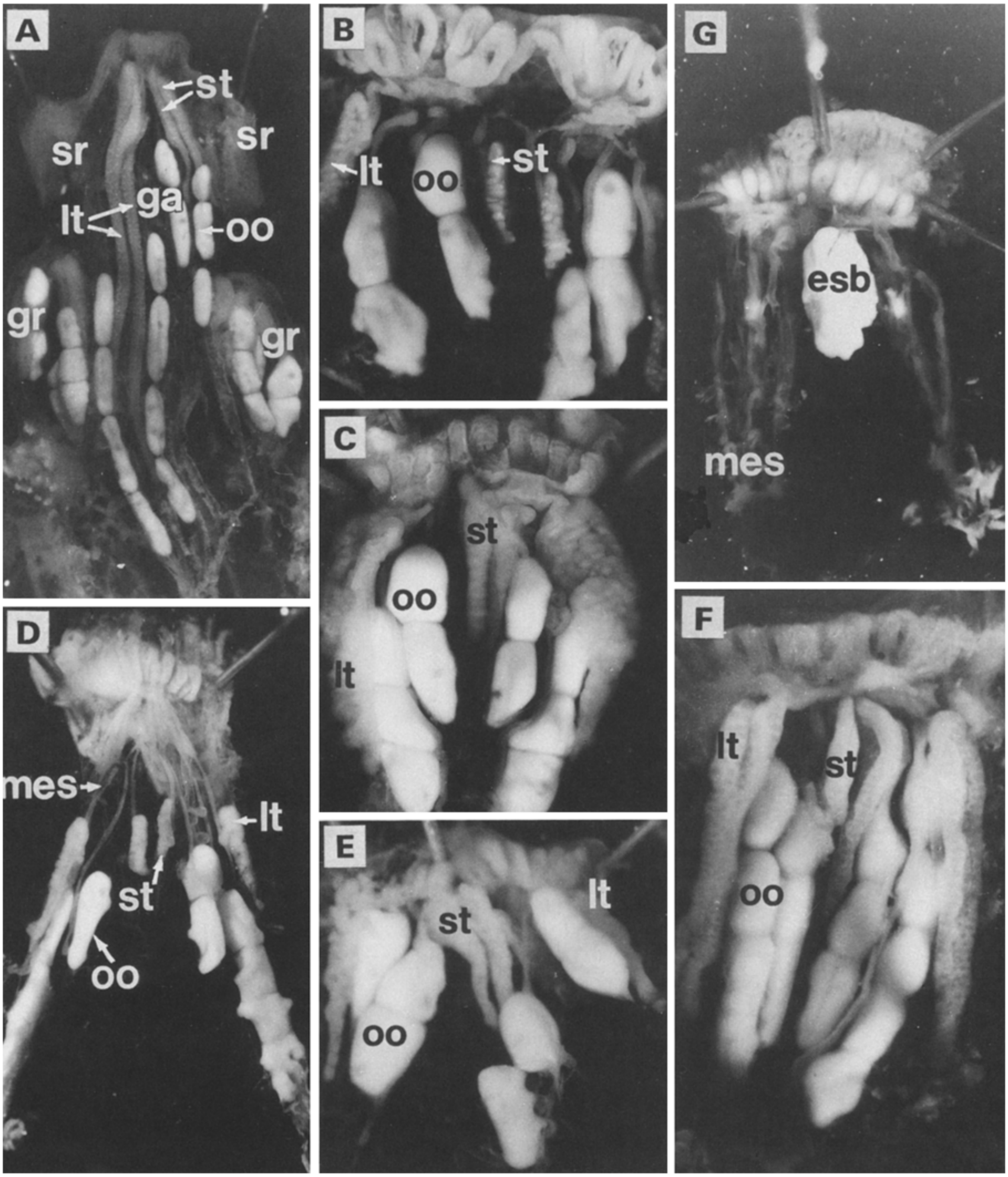

Fig. 4. *Acropora* spp. Dissections showing mature gonads in various species. (A) Branchlet of *A. granulosa* showing sterile immature radials (sr), gravid axial polyp (ga), and radial polyps (gr); the contrast between large testes (lt) and small testes (st) is pronounced in the axial polyp. (B)–(F) Radial polyps of *A. florida* (B), *A. sarmentosa* (C), *A. nobilis* (D), *A. hyacinthus* (E) and *A. longicyathus* (F), showing extended mesenteries (mes) bearing gonads; In (B)–(F), the stomodeal ring has been opened between the two inner mesenteries (Pair 3), so that large testes are displayed outermost. (G) Radial polyp of *A. loripes* fixed 4 h before spawning, showing egg-sperm bundle (esb) within the stomodeal cavity, and mesenteries extended, void of gonads. oo: oocytes

Acropora spp. polyps (both axial and radial) have twelve mesenteries, of which only eight are complete, i.e., attach to the stomodeum and bear filaments. The gonads occurred within the mesoglea of the eight complete mesenteries (Fig. 3). In mature polyps, two pairs of mesenteries bore female gonads and two bore male gonads.

The female gonads developed as single strings of oocytes in the mesoglea of the two lateral pairs of mesenteries (Pairs 1 and 2 in the developmental sequence, according to Duerden, 1902). The male gonads (here called testes) occurred as elongate cones in the mesoglea of the directive mesenteries (Pairs 3 and 4). The ventral

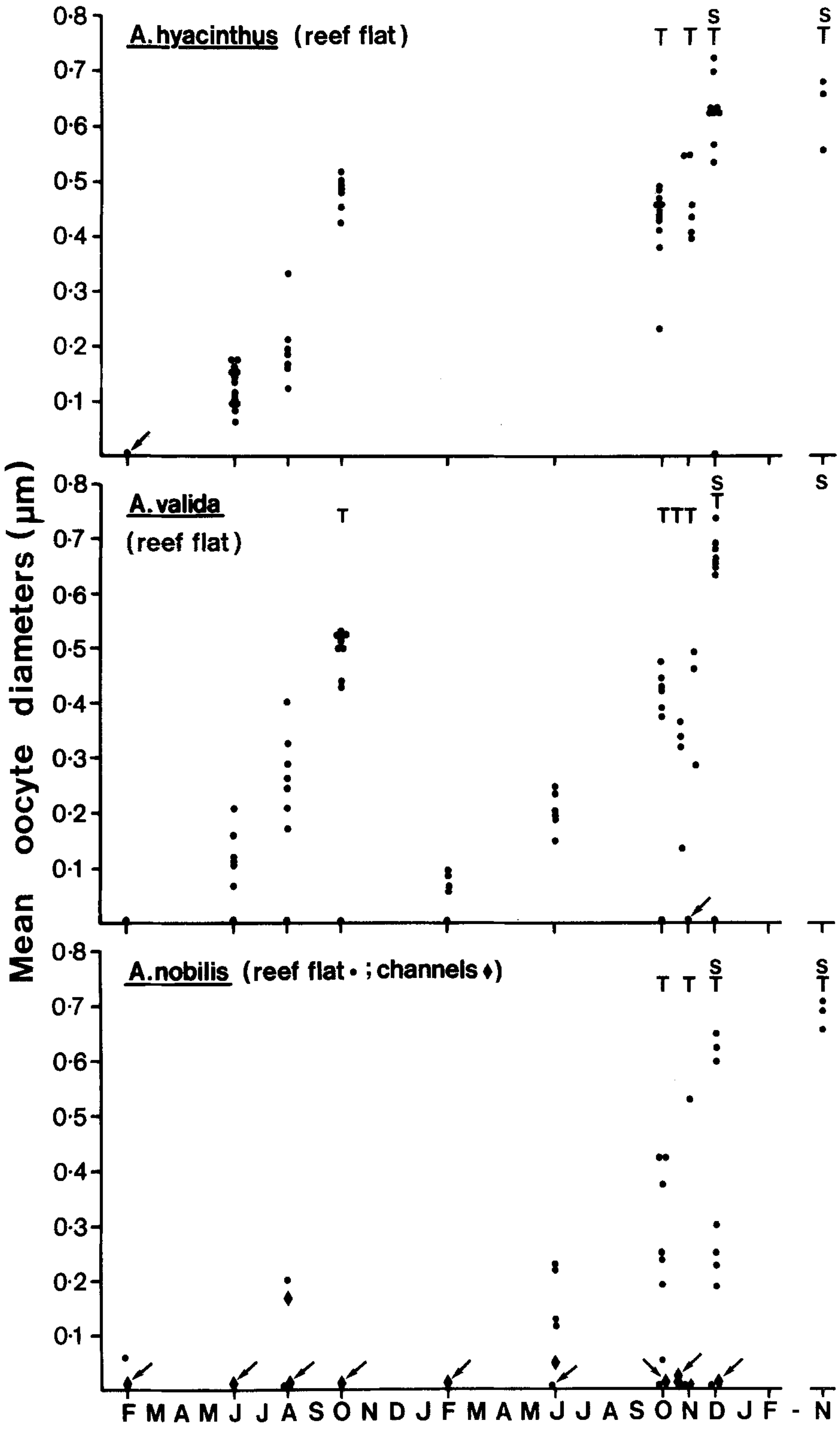

Fig. 5. *Acropora* spp. Gametogenic cycles expressed as oocyte diameters (mean for 5 polyps per colony) over 3 yr (February 1981–November 1983) for *A. hyacinthus, A. valida* and *A. nobilis*. T: testes present; S: species seen to spawn. Arrows indicate where five or more colonies were observed to have no gonads

directive mesenteries (Pair 3, or the inner pair in radial polyps) bore long testes (Fig. 4). The dorsal directives (Pair 4, outermost in radial polyps) are shorter than all other complete mesenteries, and these bore short testes. Spermatogenesis occurred within loculi in the testes (Fig. 3 B). The dorsal mesenteries and/or testes were absent from some polyps.

Gametogenic cycles

Each species had an annual gametogenic cycle in which oocytes developed over a period of about 9 mo and testes over about 10 wk (Fig. 5). Oocytes and testes matured synchronously within and between polyps within colonies, and all or most of the reproductive products of a colony

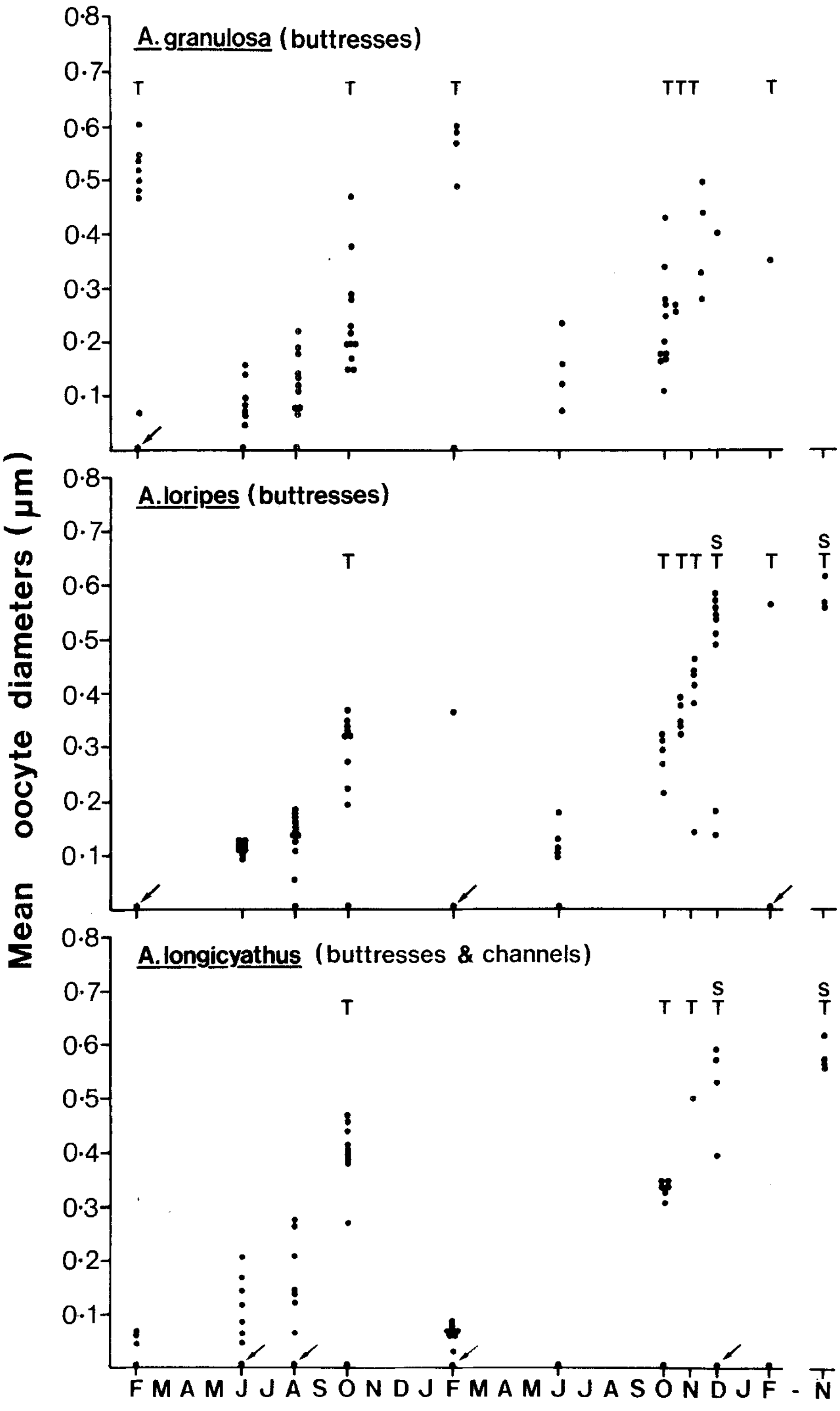

Fig. 6. *Acropora* spp. Gametogenic cycles for *A. granulosa, A. loripes* and *A. longicyathus*. Symbols as in Fig. 5

were expelled on a single night of the year. The commonest pattern (seen in *Acropora loripes, A. longicyathus, A. florida, A. nobilis, A. hyacinthus, A. valida*) was for gametes to mature in the late spring-early summer.

Gametogenesis was synchronous throughout the population for *Acropora hyacinthus* (Fig. 5), and almost so for *A. loripes* (Fig. 6). Oocyte development began some time between February and June for these species, and gamete release occurred after the full moon of late November or early December, when sea temperatures were 28° to 29 °C. A similar pattern occurred for *A. nobilis* and *A. valida* (Fig. 5), except that these species probably had a split spawning season, with some colonies spawning before or after the major spawning event. This was inferred from the presence of two categories of oocyte size in *A. nobilis* (Fig. 5) and the presence of some colonies of both species

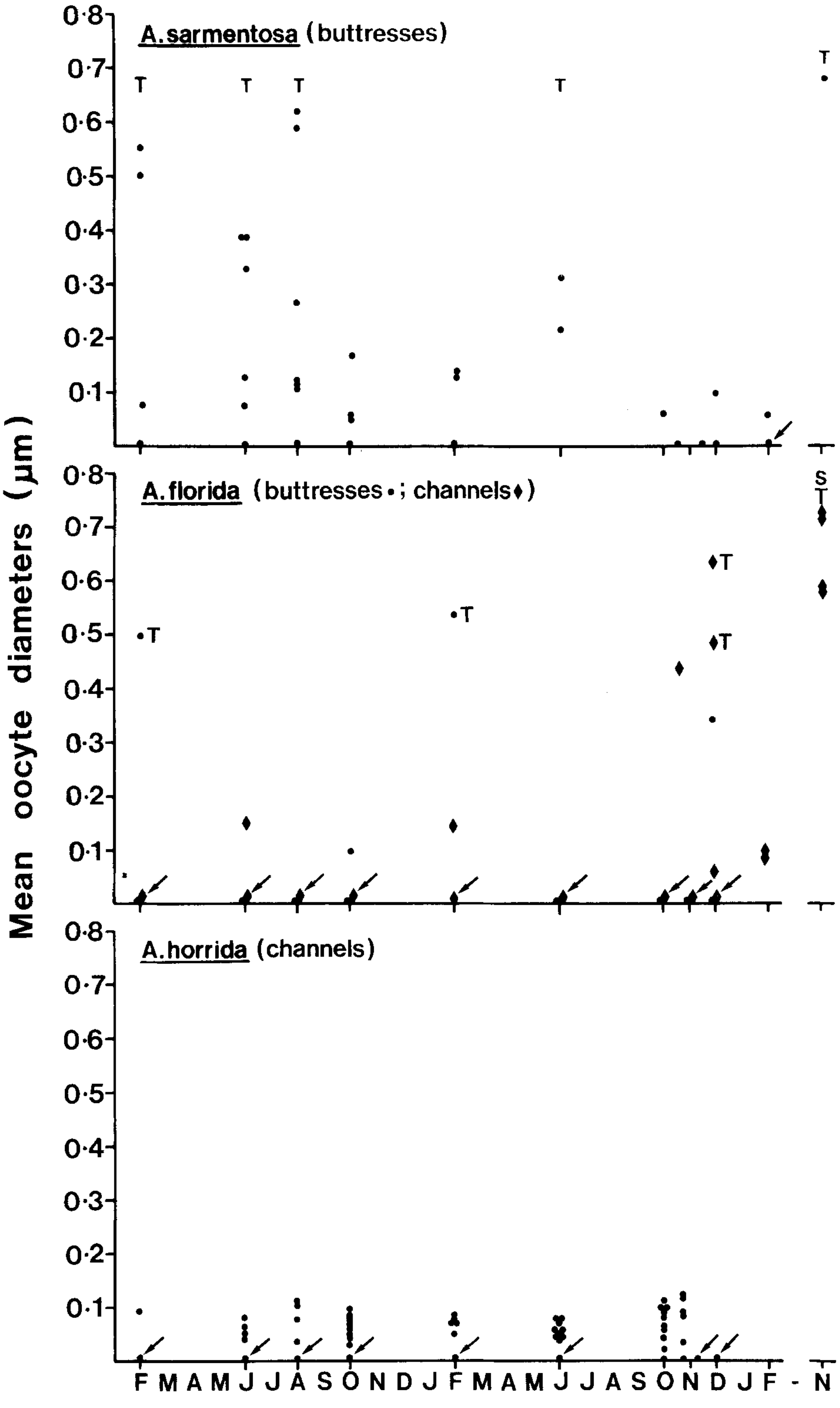

Fig. 7. *Acropora* spp. Gametogenic cycles for *A. sarmentosa*, *A. florida* and *A. horrida*. Symbols as in Fig. 5

which had apparently voided their reproductive products before the major spawning event. *A. longicycathus* also released gametes in the major spawning event, but its oogenic cycle was longer, oocyte development being clearly visible by February (Fig. 6). Records of gametogenesis for *A. florida* are meagre, because samples taken early in the study proved to be too short. The data obtained indicate that most of the colonies in the population spawned during the mass spawning event of early summer, with a few maturing in late summer (Fig. 7).

Two species, *Acropora granulosa* and *A. horrida*, did not spawn during the major spawning event. *A. granulosa* matured three months later: coloured eggs and motile sperm were present in early February for this species (Fig. 6), indicating a spawning time of late February or early March (i.e., late summer). None of the *A. horrida*

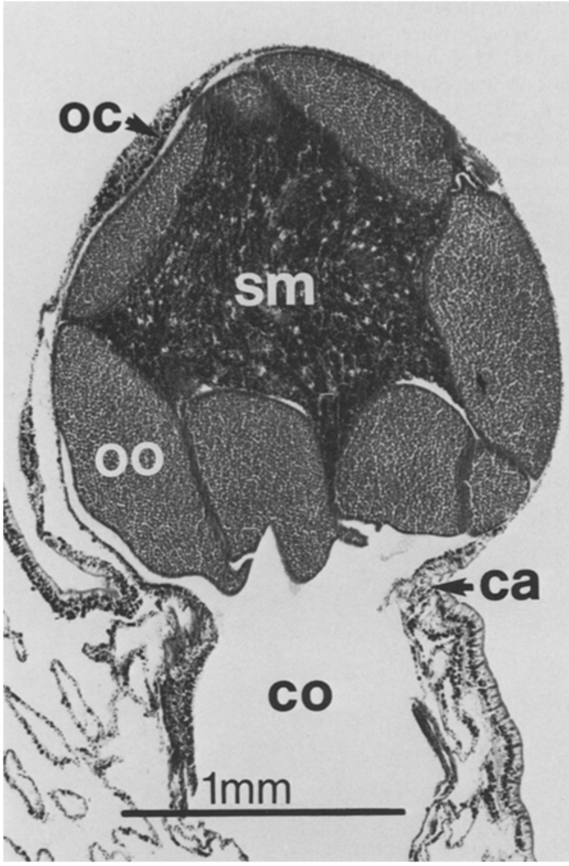

Fig. 8. *Acropora valida.* Longitudinal section of radial polyp in "setting" stage, minutes before spawning. The egg-sperm bundle is held under the oral cone (oc), which is distended beyond the calice (ca). Eggs (oo) surround a sperm mass (sm), which does not have the loculated appearance as in mature testes. co: coelenteron

colonies sampled were ever seen to have mature gametes (Fig. 7), although immature oocytes could be found in colonies at any time of the year. All colonies sampled were in channel sites. Towards the end of the study period, colonies of *A. horrida* were sought from shallower locations outside the quadrats. These were found to have maturing oocytes and very immature testes in late November, suggesting a spawning time in late summer. *A. sarmentosa* did not show the synchrony between colonies seen in other species. Different colonies of this species were found to be reproductively mature in February, August and November (Fig. 7).

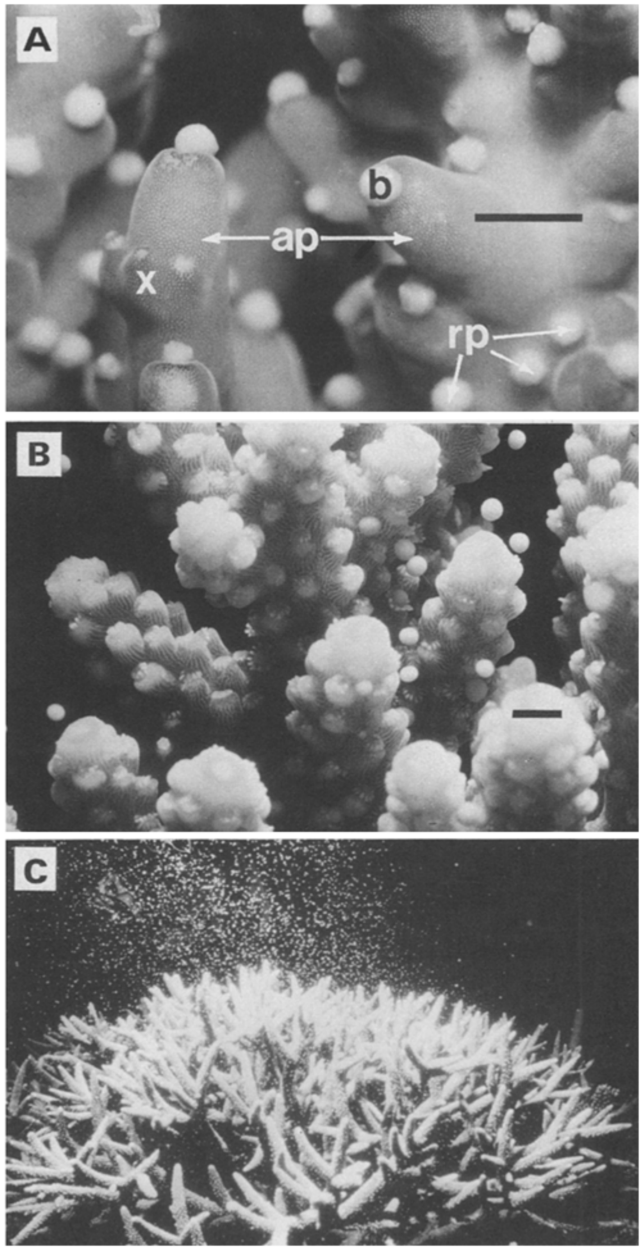

Fig. 9. *Acropora* spp. Spawning behavior in three species. (A) "Setting" of egg-sperm bundles (b) prior to release in *A. loripes;* axial polyps (ap) and most radial polyps (rp) are fertile; incompletely formed radial polyps (x) are infertile. (B) Release of egg-sperm bundles in *A. florida;* only radial polyps at bases of branchlets are fertile. (C) Synchronous release of egg-sperm bundles by colony of *A. nobilis.* Scale bars = 5 mm

Spawning behaviour and participation in the mass spawning event

In all species observed to spawn (i.e., all species except *Acropora granulosa* and *A. horrida*), each gravid polyp spawned gametes in an "egg-sperm bundle", consisting of all its eggs and testes (Fig. 8). This bundle was formed in the polyp cavity some hours before spawning (Fig. 4 G). Within an hour before spawning, colonies could be seen to be "setting", i.e., the circum-oral area was distorted by the presence of an egg-sperm bundle behind it (Fig. 8). The egg-sperm bundle was then slowly extruded through the mouth, and after release floated rapidly to the water surface (Fig. 9). Six species, *A. loripes, A. longicyathus, A. florida, A. nobilis, A. hyacinthus* and *A. valida* spawned in this manner on the same evening (26 November) in 1983 (Table 4). Thirteen other species of *Acropora* were observed to spawn at this time, and a further nine with

Table 4. *Acropora* spp. Timing of reproductive maturation and spawning in 41 species at study sites on Big Broadhurst Reef, with confirmatory observations at nearby Bowden Reef. Nomenclature from Veron and Wallace (1984). Asterisks indicate species from present study. Maturation: M, gonads mature November 26, 1983 (eggs coloured, sperm motile); x, gonads immature or not present, November 26, 1983. Spawning: f, (f) spawning seen, (spawning inferred) in field, December 6, 1982; A, spawning seen in aquaria, November 26, 1983; F, spawning seen in field, Bowden Reef, November 26, 1983. Absence of samples from either reef slope or reef flat indicates the species did not occur there

Species	Maturation						Spawning		
	Reef slope		Reef flat		Total				
A. secale	4M	3x			4M	3x		A	F
*A. loripes**	15M				15M		f	A	
*A. granulosa**		9x				9x			
*A. florida**	3M	3x	4M	1x	7M	4x		A	
*A. sarmentosa**	1M	2x			1M	2x			
A. dendrum		1x				1x			
A. latistella	3M	2x			3M	2x			
*A. longicyathus**	8M				8M			A	
A. tenuis	1M	3x			1M	3x		A	
A. elseyi	5M				5M			A	
A. cerealis	3M	3x			3M	3x		A	F
A. carduus		4x				4x			
A. selago	1M	1x			1M	1x			
A. clathrata	5M				5M		(f)		
A. subulata	1M				1M				
A. aculeus		1x				1x			F
A. formosa	15M	8x			15M	8x			
A. samoensis	3M	1x	1M		4M	1x			
A. divaricata	4M	3x		1x	4M	4x			
*A. horrida**		8x				8x			
A. yongei		4x	3M		3M	4x		A	
*A. nobilis**	8M	7x	8M	1x	16M	8x	(f)		F
A. valenciennesi	2M	3x	1M		3M	3x			
A. millepora	1M		3M	2x	4M	2x	(f)	A	F
A. lutkeni	4M		9M	5x	13M	5x		A	
*A. hyacinthus**	10M		8M	1x	18M	1x	(f)	A	F
A. cytherea	2M	1x	1M	1x	3M	2x	(f)	A	
A. nasuta	1M		7M		8M		f	A	F
A. austera			4M		4M		(f)	A	
A. gemmifera			9M	1x	9M	1x	f	A	F
A. monticulosa			5M	1x	5M	1x			
A. digitifera				6x		6x			
A. aspera				1x		1x			
A. humilis			10M	1x	10M	1x	f	A	
A. anthocercis			1M		1M				
A. nana				1x		1x			
A. robusta			6M		6M			A	
*A. valida**			4M	2x	4M	2x	(f)	A	F
A. grandis				2x		2x			
A. multiacuta				5x		5x			
A. microphthalma							(f)		

mature oocytes and testes may have spawned also (Table 4). *A. granulosa* did not spawn during the mass spawning event, nor evidently did *A. horrida.* One colony of *A. sarmentosa* was seen to spawn; others, from their gametogenic cycle, clearly spawned at other times of the year. All or most of the colonies in the population spawned on the same night in *A. loripes, A. longicyathus* and *A. hyacinthus.* Some colonies of *A. nobilis* and *A. valida* had probably spawned the previous month, and the proportion of the population spawning together in *A. florida* is uncertain.

Fecundity

There was no significant change in the number of eggs per polyp from early to late oogenesis (at $p=0.05$, Student's t-test in 78 cases from 5 species). That is, there was no degeneration of developing oocytes and no sequential release of oocytes.

Fecundity of polyps (total number of eggs per polyp) varied in the following ways:

Within colonies. Polyps immediately behind the sterile zone were less fecund than other polyps in the colony

Table 5. *Acropora* spp. Polyp fecundity (no. of eggs per polyp), fecundity per cm² and geometric mean oocyte diameter over 3 yr. All measurements taken from colonies containing mature testes with motile sperm (no. of polyps per cm² counted on a 0.5 cm × 2 cm strip; mean of 3 counts per colony on 10 colonies per species). *A. horrida* was omitted because no mature colonies were found. *N:* no. of colonies sampled

Species	*N*	Polyp fecundity		Fecundity per cm²	Oocyte diam (μm)	
		$\bar{x}$ (SE)	range		Max.	Mean (± SE)
A. loripes	29	11.8 (0.67)	5–22	156	637	560 (3)
A. granulosa	38	12.8 (0.66)	6–23	116	601	534 (3)
A. sarmentosa	13	10.2 (0.66)	6–14	169	687	652 (8)
A. longicyathus	30	11.7 (0.57)	6–18	190	636	591 (5)
A. florida	21	9.5 (0.50)	6–14	209	694	584 (4)
A. nobilis	23	7.7 (0.58)	2–12	261	696	571 (6)
A. hyacinthus	41	6.1 (0.18)	4–11	182	722	622 (5)
A. valida	24	5.6 (0.37)	3–11	96	728	633 (12)

[one-way ANOVA and Student-Newman-Keuls (SNK) range test for location samples for each species]. These polyps may have been incompletely formed at the onset of gametogenesis. Beyond this "post-sterile" zone, fecundity did not vary significantly throughout the colony. The maximum length of the sterile zone plus post-sterile zone was 8 cm in *Acropora florida* (bottlebrush), 7 cm in *A. longicyathus* and *A. horrida* (both bottlebrush), 6 cm in *A. nobilis* (arborescent) 5 cm in *A. sarmentosa,* (bottlebrush), and 1.5 cm in *A. loripes* (caespitose). In *A. hyacinthus,* full fecundity was achieved on branchlets 5 cm in from the edge of the tabular colony. In *A. granulosa,* radial polyps are widely separated and it is not possible to sample five polyps at one level; usually the post-sterile zone was one or two polyps in this species. In species with bottlebrush structure, where secondary branchlets derive from larger main branches (*A. sarmentosa, A. longicyathus, A. florida* and some colonies of *A. horrida*), the post-sterile zone usually extended to the base of branchlets. The pattern in which this occurred was variable, e.g. in *A. longicyathus* branchlets at the same level could range from fully sterile to having some fully fecund polyps. This may have been due to orientation: branches of this species often lie free on the substrate, so some branchlets would be shaded and presumably growing more slowly than others.

Between colonies. All species except *Acropora hyacinthus* showed significant variability in number of eggs per polyp amongst colonies collected within one month of spawning (one-way ANOVAS at $p = 0.01$). In all species there was a continuous gradient from lower to higher values (SNK range tests).

Between species. Mean polyp fecundity ranged from 5.6 to 12.8 eggs (Table 5). There was significant variability in fecundity among species (one-way ANOVA at $p = 0.01$), and the species fell into three groups: one with high polyp fecundity (mean 11.2 eggs per polyp), *Acropora loripes, A. granulosa, A. sarmentosa, A. longicyathus* and *A. florida;* one with low polyp fecundity (mean 5.9 eggs per polyp), *A. hyacinthus* and *A. valida;* and a single species with moderately low fecundity (7.7 eggs per polyp: SNK range test), *A. nobilis.*

Between years. Of the six species tested, four (*Acropora loripes, A. granulosa, A. nobilis* and *A. hyacinthus*) showed no variation in polyp fecundity from 1981 to 1983. *A. longicythus* did vary significantly (at $p = 0.01$), having highest fecundity in 1983 (14.2 eggs per polyp) and lowest in 1982 (8.9). *A. valida,* however, showed significant increase in polyp fecundity from 4.5 in 1981 to 6.6 in 1982 (Student's t-test at $p = 0.05$). In this case, all colonies in the population were of similar size at first sampling (6.5 cm mean diam), and were probably in their first year of reproduction in 1981. By 1983, most colonies had died. One colony, 30 cm in diameter sampled in 1983, had 9.8 eggs per polyp, which was higher than in all previous samples, suggesting that this species has lowered fecundity early in life.

With polyp size. Polyp size is difficult to measure in *Acropora* spp. because polyps extend variously into and out of the branch wall, and their shapes influence the effective size of the polyp. The number of polyps per unit area of branch can be used to give an indication of effective fecundity per unit area for comparisons between species. When this was done, the estimates of relative fecundity were essentially reversed (Table 5), except for *A. valida* [but see comments on age-specific fecundity above ("Between years")].

Oocyte size and its variation

The maximum geometric mean oocyte diameter seen in the various species ranged from 601 to 728 μm and was inversely related to polyp fecundity (Table 5). All oocytes matured synchronously in polyps and in colonies in all cases, and oocyte diameter did not vary significantly within mature polyps (see Fig. 4). In five species studied in detail (*Acropora loripes, A. longicyathus, A. florida, A. nobilis* and *A. hyacinthus*), oocyte diameter (square root of maximal × medial diameter) did not vary significantly amongst locations in colonies, and oocytes were not smaller in the less-fecund post-sterile zones (one-way ANOVAs at $p=0.01$).

Age at first reproduction

Some estimate of age at first reproduction could be made for those species with more or less circular colonies when young, for colonies recruited from larvae. The smallest such colonies developing gonads (oocytes and testes) had diameters of 4 cm (*Acropora valida*), 5 cm (*A. granulosa*), 6 cm (*A. loripes*) and 7 cm (*A. hyacinthus*). For *A. valida* and *A. hyacinthus,* no smaller colonies were sampled, and all larger colonies developed gonads. These colonies were probably just over 3 yr (*A. valida*) and 4 yr (*A. hyacinthus*) old at the onset of gametogenesis, i.e., 4 and 5 yr old when they spawned, respectively. For *A. valida,* this was probably the first year of spawning, since a 2-yr old colony has only incipient branches of less than 10 mm length (Wallace, unpublished data). For *A. hyacinthus,* there may be sufficient branches and mature polyps developed by 3 yr to spawn at this age also.

In *Acropora granulosa,* four colonies of 4.5 cm to 11 cm diameter did not develop gonads, but five colonies between 5 and 10 cm diameter did, suggesting that either growth or onset of first reproduction is variable in this species. There was no suggestion of lowered fecundity in the smallest specimens: the 5 cm colony had 20 oocytes per polyp. In *A. loripes,* three colonies 4.5 to 9 cm in diameter did not develop gonads; four colonies between 6 and 10 cm did. For both these species the age of onset of reproduction is probably 4 or 5 yr. No colonies bearing only testes were found.

Discussion

Reproductive isolation through differences in timing of spawning cannot be invoked as a factor maintaining coexistence of these *Acropora* species. Most of them showed a remarkable synchronisation of gametogenesis and gamete shedding with closely related species. This synchrony extended well beyond the reef studied: three of the species and six other species of *Acropora* were seen to spawn at nearby Bowden Reef on the same night (Table 4), and four of the species plus eight other species of *Acropora* spawned on this or the previous night at Lizard Island, in the northern Great Barrier Reef (Babcock *et al.,* in press). The taxonomic extent of the synchrony was broader also. At least 20 other species from 11 genera and 4 families spawned at Big Broadhurst Reef on the same and preceding nights, and 10^{-5} scleractinian species are now known to spawn during the mass spawning event on the Great Barrier Reef (Babcock *et al.,* in press). This suggests that an extraordinary advantage is gained by spawning at the same time as other corals, and that this overrides the ecological benefits which might be obtained by partitioning reproductive times.

The proximal factors involved in synchronised spawning are suggested to be rising sea water temperature (for gamete maturation), entrainment to a lunar cue (for spawning date), then a dark period requirement (for timing of release) (Harrison *et al.,* 1984). *Acropora granulosa,* one of the two species not spawning during the mass spawning event, may be sheltered by its preference for deeper, shaded positions (Wallace, 1978) from these cues. The other species, *A. horrida,* did not complete gametogenesis in the surge channels, which are its favoured habitat (Wallace, 1978). Some recruitment of this species from larvae occurred (Table 2), showing that it did reproduce sexually somewhere, but its gametogenic cycle elsewhere on the reef suggested that its maturation and spawning occurred well after early summer. Colonies of this species in which gametogenesis did not proceed to completion differed from those in which it did in two ways: they were in deeper water and they underwent chronic fragmentation and partial burial. It is possible that their resources were at all times allocated to extension in preference to sexual reproduction. A third species, *A. sarmentosa,* had some colonies that matured to spawn in the mass spawning event, but there appeared to be little synchrony in maturation of colonies of this species.

If reproductive isolation is not occurring for most species, what factors in the biology of the species may be contributing to the coexistence of so many species within easy reproductive range? The fact that they spawn buoyant gametes, which must undergo fertilization and a period of embryogenesis and maturation in the water column before settling out of the plankton, strongly suggests that offspring are carried well away from the parent population. This is supported by the finding that larval recruitment occurs independently of the presence of adult colonies (Table 2). In contrast to the situation for larval brooding species and species with solely benthic larvae (Loya, 1976; Fadlallah, 1983), sexual reproduction by the resident population is unlikely to directly influence population structure or to be responsible for the local persistence of the species. Rather, this function will be served by recruitment of immigrant larvae and of locally produced fragments.

Since recruitment by larvae is independent of the prior presence of a species, it provides a means by which the possibility of establishment of the species is renewed annually. Great potential for variability exists within this process, which is both differential and temporally variable. It is a seasonal event, which occurs mainly over a brief summer period (Wallace and Bull, 1982; Wallace, 1983 b, 1985), so variability in environmental factors at this time of the year could be expected to affect both availability and survival of settling planulae (Jackson and Strathman, 1981), and possibly also the relative settlement success of different species in different years. Overall intensity of recruitment has also been found to vary significantly from year to year during a four-year study at Big Broadhurst Reef (Wallace 1983 a, 1985, and unpublished data). Thus, year groups will differ significantly in numbers, and this may influence the way that various biological factors affect subsequent survival. For example, members of some large annual cohorts may have a chance to escape predation by virtue of their numbers (Underwood and Denley, 1984), and competitive events may be more or less significant in some years.

Recruitment by larvae was found to favour certain species which seldom or never recruit by fragments (Table 2). These species will thus be more evenly distributed as juveniles than fragmenting species, a factor which may be later accentuated by subsequent fragmentation of sparsely distributed species. Some species recruiting densely from larvae and surviving to visible size may suffer subsequent mortality a considerable time after the initial recruitment process. Heavy larval recruitment of *Acropora hyacinthus* to all sites where adults were rare (Table 2) suggests (without proving) that these recruits will suffer heavy mortality. This species may be structurally unsuited to colony development in either sandy or steeply sloping habitats (buttresses or channels), because of its need to form a strong attachment to support the table-shaped colony which develops after about 4 yr (Wallace, 1978).

Subsequent to larval recruitment and survival to a size at which branches can be broken off, some species reproduce asexually by fragmentation. For fragmenting species, introduction of even a single larval recruit into a habitat favourable to survival of fragments will have a profound effect. This, with sparse larval recruitment, may explain why whole surge channels may be found without colonies of the fragmenting species *Acropora longicyathus* or *A. horrida,* while others are dominated by one or other or both of these species. The success of recruitment by fragmentation is mediated by habitat type. Some species, e.g. *A. longicyathus* and *A. horrida,* have poor cementing qualities, fragments always lying free on the substratum, while others, such as *A. florida* and *A. nobilis,* rapidly produce large quantities of coenosteum and cement themselves to the substratum. The former will be unsuccessful at colony multiplication in habitats with strong scouring water motion. Fragmentation also occurs in response to different factors in different species, e.g. *A. florida,* a sturdy species, was only observed to fragment in storms, while *A. longicyathus* and *A. horrida* fragmented year round. *A. florida* was also a "moderately good" larval recruiter, as might be expected by its position halfway between "fragmenters" and "non fragmenters".

Oocyte size, fecundity and mode of spawning

Compared with other corals whose sexual characteristics have been reported, these species of *Acropora* have very large eggs and low polyp fecundity (e.g. see tables in Rinkevich and Loya, 1979; Harriott, 1983). The oocytes of many corals including *Acropora* are oval or irregular in shape, and mature oocytes may be more or less elongate when fixed within the polyps (e.g. compare the fixed dimensions of oocytes in axial and in radial polyps of *A. granulosa* in Fig. 4 A). Most records of oocyte size, being taken from histological sections, quote the longest measured dimension (e.g. Rinkevich and Loya, 1979; Fisk, 1981; Fadlallah and Pearse, 1982; Harriott, 1983; Babcock, 1984), and the longest observed to date has been 1 200 μm in the solitary coral *Balanophyllia elegans* (Fadlallah and Pearse, 1982). The maximum geometric mean diameter measured for *Acropora* spp. in the present study was 728 μm (Table 5), but the maximal diameters ranged up to 1 600 μm, making these oocytes longer than any previously recorded coral oocytes.

The lowest mean fecundity (5.6 eggs per polyp) approaches that of *Stylophora pistillata* (up to 5), a species shown to have "*r* strategist" characteristics (Loya, 1976).

Two features of variation in fecundity are obvious from this study:

Its relationship to age and to growth. Polyps in actively extending axial polyps and newly formed radial polyps do not develop gonads; immature radial polyps and, at least in some species, polyps in submature colonies, are not fully fecund. The sequence of gamete maturation in *Acropora* spp. polyps, in which female gametes requiring long maturation time develop on the first-formed mesenteries, and testes with short maturation times develop on the last-formed mesenteries, may provide a means by which reproduction can be achieved even in polyps which are incompletely formed at the onset of gametogenesis.

Its relationship to habitat. Failure to form gametes, or failure of gametes to mature, in two species in channels could result from sub-optimal conditions in these locations. It might also again be related to growth: some species of *Acropora* extend more rapidly at deep sites than at shallow sites (Gladfelter *et al.,* 1978; Oliver *et al.,* 1983). The two species involved (*Acropora horrida* and *A. nobilis*) were of the rapidly extending arborescent type.

Clearly, fecundity can vary from place to place, and during the lifetime of a colony. It should also follow that polyp fecundity may be reduced in some years by adverse environmental conditions. There was some suggestion of this for *Acropora longicyathus* in this study, but not for other species.

Various authors have sought to categorize corals in terms of their modes of spawning. Stimson (1978) suggested that different modes of reproduction may be employed by deep and shallow-water corals. This was based on observations of planulation in seven species of shallow-water corals, including three species of *Acropora,* and failure to detect any spawning in three species of *Acropora* from deep water or occurring over a wide range of depths. Of the species of *Acropora* observed to planulate, one, *A. humilis* has been seen to release gametes on the Great Barrier Reef (present Table 4; Bothwell, 1982; Harrison *et al.,* 1984). It thus appears that different modes of spawning may be employed by the same species in different locations. However, the method used by Stimson (1978), where unfiltered seawater from holding tanks was examined, does not preclude the possibility that the planulae seen were from other corals.

Other authors have sought to identify aspects of the spawning process which might limit corals to particular distribution patterns. Kojis and Quinn (1981), from observations of sticky, negatively buoyant eggs in *Goniastrea favulus* (as *G. australiensis*), proposed that "species characteristic of reef flats ... may have various modes of reproduction that reduce planktonic life to a minimum and retain larvae on the reef flat.". Babcock (1984) observed that a closely related species, *G. aspera,* has eggs which are positively buoyant and quickly dispersed, but he failed to find any difference in the degree of aggregation of the two species on the reef flat.

The results of the present study suggest that reef-flat species of *Acropora* (*Acropora*) have fewer, larger eggs than reef-slope species (but note that one of the reef-flat species *recruits* heavily on the slope). This is not related to polyp size, and in some species with closely arranged polyps the fecundity of whole colonies is very high. *A. sarmentosa,* distributed through all habitats, has intermediate egg size and fecundity, and seems to confirm this observation. The significance of larger egg size is not immediately obvious, other than in the possibility that larger eggs may carry greater nutrient stores, and thus may result in longer-lived planulae. If, for example, larger planulae were more buoyant, those planulae might settle out only on contact with shallow substratum, thus being confined to shallow habitats. The difference in size between reef-flat and reef-slope species is being studied further with samples from a larger set of species, collected when gonads were mature in 1983.

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